

TASK 2 – SENSOR BASED SIMULATION

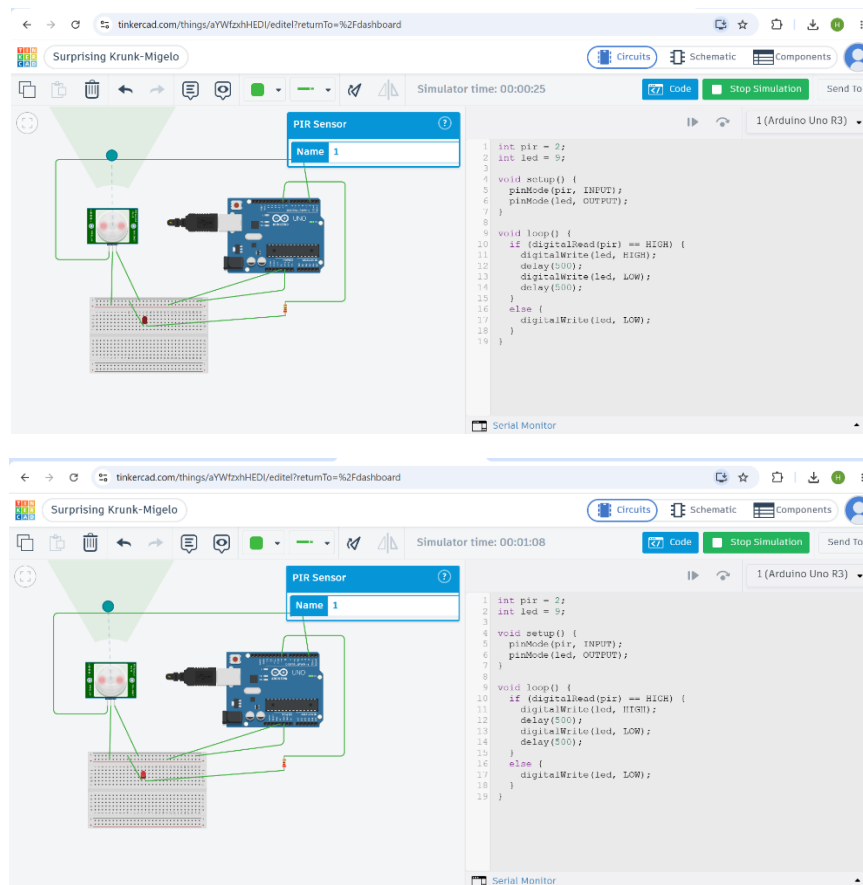


Fig : Simulation

Code Explanation

The program is used to control an LED based on motion detected by a PIR sensor.

1. Pin Declaration

```
int pir = 2;
```

```
int led = 9;
```

Here, the PIR sensor is connected to digital pin 2 and the LED is connected to digital pin 9.

2. Setup Function

```
void setup() {
  pinMode(pir, INPUT);
```

```
pinMode(led, OUTPUT);  
}
```

The `setup()` function runs once when the Arduino starts. The PIR pin is configured as an input to receive the sensor signal, and the LED pin is configured as an output to control the LED.

3. Loop Function

```
void loop() {
```

The `loop()` function runs continuously and checks the PIR sensor repeatedly.

4. Motion Detection

```
if (digitalRead(pir) == HIGH) {
```

The `digitalRead()` function reads the PIR sensor. If it gives a HIGH signal, motion is detected.

5. LED Control

```
digitalWrite(led, HIGH);
```

```
delay(500);
```

```
digitalWrite(led, LOW);
```

```
delay(500);
```

When motion is detected, the LED turns ON for 500 milliseconds and then turns OFF for 500 milliseconds. This makes the LED blink.

6. No Motion

```
else {
```

```
    digitalWrite(led, LOW);
```

```
}
```

If no motion is detected, the LED remains OFF.

Working Principle

The PIR sensor detects motion and sends a HIGH signal to the Arduino, which makes the LED blink. When there is no motion, the PIR sends a LOW signal and the LED remains OFF.

Conclusion

The project successfully demonstrates motion detection using a PIR sensor and Arduino. The LED provides a simple visual indication when motion is detected. This system can be useful for basic security and automatic lighting applications.